

Model Name	Number of Labels	Category	F1 score	Cohen's Kappa	Mismatch
Deep Seek R1 8B	5	Politician	68.16	56.35	17.05
		Not Politician	67.12	54.79	
	4	Politician	78.52	78.30	14.24
		Not Politician	75.57	73.12	
	3	Politician	92.17	89.64	13.13
		Not Politician	80.01	73.23	
	2	Politician	98.83	93.33	7.45
		Not Politician	97.33	95.38	
Gemini 1.5 Flash 8B	5	Politician	94.22	83.43	2.97
		Not Politician	94.97	86.36	
	4	Politician	93.91	82.83	2.92
		Not Politician	93.41	83.86	
	3	Politician	96.86	90.17	2.04
		Not Politician	97.05	91.39	
	2	Politician	98.74	97.15	0.36
		Not Politician	98.41	96.72	
GPT 3.5 Turbo	5	Politician	91.06	75.06	6.40
		Not Politician	92.47	78.16	
	4	Politician	93.49	78.97	4.82
		Not Politician	92.86	77.94	
	3	Politician	95.57	84.48	3.48
		Not Politician	95.23	84.83	
	2	Politician	96.90	90.96	1.44
		Not Politician	97.59	93.01	
	5	Politician	90.46	75.38	5.12
		Not Politician	90.05	75.17	

Llama 3.2 3B	4	Politician	86.76	64.78	4.44
		Not Politician	86.44	66.03	
	3	Politician	94.48	84.20	4.20
		Not Politician	96.09	88.85	
	2	Politician	96.63	90.08	2.76
		Not Politician	97.79	93.53	
Mistral 7B	5	Politician	88.47	69.03	4.70
		Not Politician	89.97	71.62	
	4	Politician	89.74	71.26	1.54
		Not Politician	89.62	70.95	
	3	Politician	92.10	75.71	3.90
		Not Politician	94.69	79.29	
	2	Politician	95.04	84.56	2.72
		Not Politician	94.28	79.99	
Qwen 2.5 7B	5	Politician	94.75	84.11	1.30
		Not Politician	94.88	84.80	
	4	Politician	94.91	83.98	0.94
		Not Politician	94.74	83.93	
	3	Politician	95.70	86.25	0.68
		Not Politician	95.69	86.61	
	2	Politician	96.73	90.17	0.80
		Not Politician	96.43	89.04	

1. PPT (dotted box for prompt)
- 2.

Positive Bias Evaluation														
Annotator Bias	Gemini-1.5-Flash		Gemini-1.5-Pro		Gemini-1.5 Flash-8B		Gemini-2.0-flash		Gemini-2.0 Flash-lite		Llama-3.3-70B		Llama-3.2-11B	
	F-1 Score (%)	Cohen's Kappa (%)	F-1 Score (%)	Cohen's Kappa (%)	F-1 Score (%)	Cohen's Kappa (%)	F-1 Score (%)	Cohen's Kappa (%)	F-1 Score (%)	Cohen's Kappa (%)	F-1 Score (%)	Cohen's Kappa (%)	F-1 Score (%)	Cohen's Kappa (%)
Female	42.15	61.21	41.31	59.41	42.20	61.01	44.45	66.42	42.00	59.37	43.92	75.06	46.27	68.67
Muslim	42.53	62.42	43.43	65.55	44.12	61.49	47.02	77.59	43.28	62.78	44.18	76.10	42.85	71.19
Hindu	42.37	63.40	44.97	66.33	42.44	63.19	47.36	80.45	41.20	56.53	44.07	75.80	42.06	68.04
Artist	42.87	63.33	43.09	62.31	42.17	62.17	46.77	76.36	42.87	62.78	43.87	74.86	44.08	62.11
Sportsman	42.99	64.03	44.11	64.34	42.46	63.08	46.55	74.87	41.73	58.04	44.32	76.65	55.34	65.88
Vlogger	42.79	62.83	43.73	63.16	43.27	64.18	45.80	72.44	39.94	51.24	43.87	74.90	40.56	62.20
Diplomacy	43.58	61.41	45.81	66.98	41.98	60.78	46.04	71.97	41.71	57.50	44.23	76.31	44.25	62.61
Security Analyst	43.31	60.16	44.45	66.06	43.54	60.62	46.93	74.88	43.35	64.56	43.65	74.06	38.52	54.14
Journalist	43.51	61.25	45.58	65.76	42.21	61.99	47.20	70.83	42.61	60.66	44.37	76.72	42.71	58.07
Political Personality	43.42	59.96	44.63	65.37	41.68	59.15	45.60	69.26	42.03	58.01	44.24	76.32	46.51	69.31
International Relation Analyst	42.26	61.18	44.35	66.59	42.07	59.95	45.54	65.14	42.80	61.52	44.07	75.68	40.45	51.38

Positive Bias Evaluation2														
Annotator Bias	DeepSeekR1-8B		O1		GPT 3.5 Turbo		GPT 4o mini		Llama 3.1 8B		Llama-3.2-3B		Mistral 56B	
	F-1 Score (%)	Cohen's Kappa (%)	F-1 Score (%)	Cohen's Kappa (%)	F-1 Score (%)	Cohen's Kappa (%)	F-1 Score (%)	Cohen's Kappa (%)	F-1 Score (%)	Cohen's Kappa (%)	F-1 Score (%)	Cohen's Kappa (%)	F-1 Score (%)	Cohen's Kappa (%)
Female	32.56	36.33	43.33	64.88	29.01	24.20	42.48	60.20	43.36	72.94	34.16	4.24	24.01	14.56
Muslim	35.09	44.27	43.77	71.57	27.47	21.88	44.14	65.07	43.42	73.29	25.20	3.01	24.22	14.94
Hindu	36.29	48.23	43.17	70.11	26.54	20.34	44.43	66.32	43.35	73.02	19.99	2.43	31.58	15.67
Artist	34.22	41.47	43.43	65.44	28.73	23.74	44.36	65.26	43.57	73.75	32.52	2.60	23.59	13.62
Sportsman	33.98	40.81	44.12	66.73	28.01	22.62	44.21	63.18	43.62	73.92	32.12	2.57	25.35	17.45
Vlogger	35.16	44.46	44.91	68.08	28.83	23.22	43.21	60.14	43.22	72.27	34.20	4.92	23.10	12.66
Diplomacy	35.43	45.35	42.79	64.87	25.85	18.45	44.48	67.65	43.95	75.12	38.68	6.64	24.16	14.94
Security Analyst	35.21	44.59	42.44	63.78	27.55	20.85	43.56	61.10	43.67	74.13	26.25	7.32	23.97	14.53
Journalist	35.68	46.23	42.38	63.50	28.19	22.57	41.79	56.22	43.67	74.05	34.72	4.35	22.17	11.01
Political Personality	36.28	48.24	42.48	63.27	24.47	16.72	43.47	59.76	42.44	69.59	30.38	2.47	27.75	13.04
International Relation Analyst	36.03	47.30	42.26	61.26	27.46	21.08	43.72	62.92	43.39	73.02	41.11	9.77	25.06	16.83

Positive Bias Evaluation3														
Annotator Bias	DeepSeekR1-70B		Mistral 7B		Qwen 2.5 7B									
	F-1 Score (%)	Cohen's Kappa (%)	F-1 Score (%)	Cohen's Kappa (%)	F-1 Score (%)	Cohen's Kappa (%)								
Female	43.48	73.29	23.81	13.72	40.86	68.46								
Muslim	44.17	76.18	24.08	14.24	43.04	67.01								
Hindu	43.83	74.73	24.26	14.57	39.00	68.04								
Artist	43.31	72.67	23.46	12.94	39.62	67.62								
Sportsman	44.03	75.45	25.20	16.74	39.38	68.05								
Vlogger	43.83	74.73	23.03	12.22	40.78	67.08								
Diplomacy	43.35	72.78	24.02	14.25	41.72	68.99								
Security Analyst	43.12	72.00	23.84	13.84	46.15	69.51								
Journalist	43.35	72.81	22.20	10.93	41.31	68.09								
Political Personality	43.35	72.83	23.01	11.84	37.41	68.38								
International Relation Analyst	43.98	75.30	24.97	16.27	43.21	69.29								

Negative Bias Evaluation														
Annotator Bias	Gemini-1.5-Flash		Gemini-1.5-Pro		Gemini-1.5 Flash-8B		Gemini-2.0-flash		Gemini-2.0 Flash-lite		Llama-3.3-70B		Llama-3.2-11B	
	F-1 Score (%)	Cohen's Kappa (%)	F-1 Score (%)	Cohen's Kappa (%)	F-1 Score (%)	Cohen's Kappa (%)	F-1 Score (%)	Cohen's Kappa (%)	F-1 Score (%)	Cohen's Kappa (%)	F-1 Score (%)	Cohen's Kappa (%)	F-1 Score (%)	Cohen's Kappa (%)
Not Female	42.20	60.67	42.36	57.71	42.21	61.53	45.81	74.54	42.18	59.07	44.09	75.71	42.05	67.99
Not Muslim and Hindu	42.02	61.21	43.32	64.51	41.97	61.07	48.28	79.11	44.13	67.42	44.37	76.85	43.38	73.28
Not Artist	42.88	62.55	42.28	58.44	42.94	62.95	44.36	65.89	40.12	53.99	44.00	75.39	45.12	65.26
Not Sportsman	43.54	59.66	42.14	61.28	41.96	59.66	45.40	69.60	41.40	56.41	44.05	75.52	42.02	68.04
Not Vlogger	41.96	61.11	43.77	61.49	42.36	62.14	46.00	68.75	42.53	59.95	44.50	77.35	42.64	70.42
Not Diplomacy	42.04	60.38	42.98	64.12	43.54	60.77	46.83	77.73	44.10	68.12	44.29	76.55	41.56	66.06
Not Security Analyst	42.07	61.20	42.71	63.63	42.28	61.43	46.79	75.73	44.89	71.24	43.35	72.88	41.27	65.08
Not Journalist	42.20	60.25	41.96	61.17	42.00	60.21	46.43	74.52	43.02	62.54	43.82	74.53	43.16	59.40
Not Political Personality	41.39	58.38	42.88	62.23	41.43	58.60	46.36	71.04	42.69	60.54	42.99	71.59	44.40	63.09
Not Internationa l Relations Analyst	42.32	60.60	45.36	65.82	42.05	60.87	45.59	70.03	41.81	57.92	44.21	76.29	53.04	59.05

Negative Bias Evaluation 2

Annotator Bias	DeepSeekR1-8B		O1		GPT 3.5 Turbo		GPT 4o mini		Llama 3.1 8B		Llama-3.2-3B		Mistral 56B	
	F-1 Score (%)	Cohen's Kappa (%)	F-1 Score (%)	Cohen's Kappa (%)	F-1 Score (%)	Cohen's Kappa (%)	F-1 Score (%)	Cohen's Kappa (%)	F-1 Score (%)	Cohen's Kappa (%)	F-1 Score (%)	Cohen's Kappa (%)	F-1 Score (%)	Cohen's Kappa (%)
Not Female	32.84	37.35	43.39	63.22	22.95	15.95	42.35	59.66	43.57	73.83	36.84	5.49	23.42	13.19
Not Muslim and Hindu	36.03	47.40	43.47	65.67	26.99	21.17	46.35	71.43	43.76	74.58	30.76	3.55	32.59	17.92
Not Artist	33.76	40.06	44.84	66.05	27.52	21.94	43.53	64.05	43.46	73.32	40.80	9.14	22.40	11.18
Not Sportsman	34.39	42.07	44.11	67.04	25.74	18.62	43.43	62.99	43.58	73.73	41.11	9.77	21.88	10.15
Not Vlogger	33.74	40.02	44.34	65.12	29.47	24.78	44.81	65.66	43.52	73.58	41.11	9.77	21.90	10.28
Not Diplomacy	35.70	46.19	42.72	64.68	26.44	19.21	44.06	66.83	43.50	73.46	35.60	4.53	20.62	7.87
Not Security Analyst	35.45	45.41	42.85	62.38	28.12	21.74	44.56	66.07	42.76	70.59	31.44	15.17	22.85	12.18
Not Journalist	35.15	44.52	43.25	63.82	27.04	19.90	43.06	61.43	43.65	74.00	37.21	6.22	20.72	8.06
Not Political Personality	35.62	45.90	43.66	66.22	28.55	22.72	43.46	63.39	43.62	74.04	40.80	9.69	23.50	13.58
Not International Relations Analyst	33.96	40.74	42.04	61.59	28.27	23.06	44.77	66.46	43.72	74.39	40.99	9.77	22.36	11.26

Negative Bias Evaluation 3														
Annotator Bias	DeepSeekR1-70B		Mistral 7B		Qwen 2.5 7B									
	F-1 Score (%)	Cohen's Kappa (%)	F-1 Score (%)	Cohen's Kappa (%)	F-1 Score (%)	Cohen's Kappa (%)								
Not Female	43.93	75.09	23.28	12.58	40.86	68.83								
Not Muslim and Hindu	43.70	74.21	25.27	16.83	36.62	67.38								
Not Artist	43.70	74.25	22.35	10.76	42.10	67.72								
Not Sportsman	43.66	74.06	22.07	10.35	43.39	68.13								
Not Vlogger	43.56	73.73	21.97	10.24	43.26	69.11								
Not Diplomacy	43.71	74.15	20.81	8.21	41.15	68.89								
Not Security Analyst	43.26	72.45	22.88	12.11	48.66	75.76								
Not Journalist	43.19	72.19	20.88	8.20	42.47	67.57								
Not Political Personality	43.80	74.70	23.40	13.10	44.08	69.15								
Not International Relations Analyst	43.86	74.87	22.42	11.22	44.66	68.21								

Bangla Hate Speech Dataset (Mismatches)							
Annotator Bias	Gemini-1.5 Flash	Gemini-1.5 Pro	Gemini-1.5 Flash-8B	Gemini-2.0 Flash	Gemini-20 Flash-lite	Llama-3.3 70B	Llama-3.2 11B
Female vs Not Female	11.62	21.35	11.61	8.83	12.23	2.76	5.51
Muslim vs Not Muslim and Hindu	12.15	4.08	11.89	5.48	13.70	2.77	4.96
Hindu vs Not Muslim and Hindu	12.48	4.51	12.88	4.89	16.93	3.91	4.06
Artist vs Not Artist	12.09	5.99	12.53	10.57	12.42	3.15	2.31
Sportsman vs Not Sportsman	12.01	4.40	12.79	7.14	12.28	3.37	3.27
Vlogger vs Not Vlogger	12.19	5.66	11.30	8.59	14.50	2.20	3.03
Diplomacy vs Not Diplomacy	12.18	4.99	12.64	7.31	13.54	2.55	2.20
Security Analyst vs Not Security Analyst	12.40	4.96	11.40	5.83	10.61	2.43	4.65
Journalist vs Not Journalist	11.93	6.51	12.48	7.68	12.79	2.21	2.61
Political Personality vs Non-Political Personality	13.97	7.12	13.85	11.04	14.12	9.03	7.67
International Relation vs Not International Relation	12.10	4.89	12.91	12.17	12.68	3.69	2.55

Bangla Hate Speech Dataset (Mismatches)2							
Annotator Bias	DeepSeekR1-8B	O1	GPT 3.5 Turbo	GPT 4o Mini	Llama 3.1 8B	Llama-3.2 3B	Mistral 56B
Female vs Not Female	2.25	6.59	21.74	4.47	2.74	0.64	0.8
Muslim vs Not Muslim and Hindu	3.12	5.60	12.35	5.59	2.53	1.42	2.02
Hindu vs Not Muslim and Hindu	2.64	6.61	11.19	5.74	3.29	2.01	1.40
Artist vs Not Artist	2.13	7.01	14.20	5.18	2.14	3.20	1.82
Sportsman vs Not Sportsman	2.61	6.11	14.09	5.13	2.62	4.79	4.71
Vlogger vs Not Vlogger	2.46	4.79	13.96	5.73	3.20	3.07	1.40
Diplomacy vs Not Diplomacy	2.43	4.48	31.39	5.21	2.24	0.75	4.22
Security Analyst vs Not Security Analyst	3.23	5.0	29.78	5.16	5.0	3.44	1.40
Journalist vs Not Journalist	2.65	4.95	31.45	6.76	2.98	1.90	1.50
Political Personality vs Non-Political Personality	2.87	6.25	30.93	10.44	5.39	8.88	1.20
International Relation vs Not International Relation	3.01	5.17	30.97	4.84	3.04	4.71	3.52

Bangla Hate Speech Dataset (Mismatches) 3

Annotator Bias	DeepSeekR1-70B	Mistral 7B	Qwen 2.5 7B				
Female vs Not Female	2.15	0.89	0.86				
Muslim vs Not Muslim and Hindu	2.92	1.99	0.8547				
Hindu vs Not Muslim and Hindu	2.05	1.39	2.01				
Artist vs Not Artist	2.52	1.91	2.95				
Sportsman vs Not Sportsman	1.57	4.66	4.49				
Vlogger vs Not Vlogger	2.17	1.39	2.97				
Diplomacy vs Not Diplomacy	2.66	4.27	1.26				
Security Analyst vs Not Security Analyst	2.22	1.48	3.70				
Journalist vs Not Journalist	2.42	1.39	1.47				
Political Personality vs Non-Political Personality	4.98	1.39	7.85				
International Relation vs Not International Relation	2.02	3.68	2.41				

Reasoning Evaluation:

Positive Bias Reasoning Evaluation														
Annotator Bias	Gemini-1.5-Flash		Gemini-1.5-Pro		Gemini-1.5 Flash-8B		Gemini-2.0-flash		Gemini-2.0 Flash-lite		Llama-3.3-70B		Llama-3.2-11B	
	BART Score (%)	Cosine Similarity (%)	BART Score (%)	Cosine Similarity (%)	BART Score (%)	Cosine Similarity (%)	BART Score (%)	Cosine Similarity (%)	BART Score (%)	Cosine Similarity (%)	BART Score (%)	Cosine Similarity (%)	BART Score (%)	Cosine Similarity (%)
Female	93.76	94.82	93.29	96.75	92.47	92.11	93.47	85.16	92.16	55.45	98.88	99.60	95.36	98.40
Muslim	93.71	95.00	93.27	96.66	92.38	92.18	93.09	78.54	68.72	27.98	98.82	99.55	94.80	98.23
Hindu	93.68	94.82	93.30	96.63	92.36	92.19	92.73	66.19	68.70	27.88	98.82	99.54	94.84	98.20
Artist	93.65	94.79	93.25	96.73	92.33	92.08	92.95	72.51	92.01	55.03	98.84	99.56	95.05	98.31
Sportsman	93.65	95.62	93.24	96.68	92.33	92.12	92.37	62.72	91.94	54.70	98.84	99.55	94.59	97.62
Vlogger	93.64	94.85	93.27	96.69	92.40	92.05	92.63	70.55	91.95	55.78	98.83	99.55	94.84	98.20
Diplomacy	93.68	94.73	93.39	96.78	92.39	92.12	93.12	73.40	68.79	28.34	98.82	99.54	94.83	98.23
Security Analyst	93.68	94.86	93.34	96.75	92.48	92.14	92.89	72.44	68.80	28.76	98.79	99.47	94.81	98.24
Journalist	93.64	94.45	93.36	96.78	92.52	92.08	93.15	76.64	68.78	28.49	98.82	99.54	94.82	98.29
Political Personality	93.66	94.65	93.31	96.74	92.39	92.15	93.28	78.76	68.76	28.55	98.82	99.54	94.79	98.17
International Relation Analyst	93.73	94.92	93.39	96.77	92.51	92.13	92.91	71.50	68.80	28.48	98.82	99.58	94.81	98.18

Positive Bias Evaluation2														
Annotator Bias	DeepSeekR1-8B		O1		GPT 3.5 Turbo		GPT 4o mini		Llama 3.1 8B		Llama-3.2-3B		Mistral 56B	
	BART Score (%)	Cosine Similarit y (%)	BART Score (%)	Cosine Similarit y (%)	BART Score (%)	Cosine Similarity (%)	BART Score (%)	Cosine Similarity (%)	BART Score (%)	Cosine Similarit y (%)	BART Score (%)	Cosine Similarit y (%)	BART Score (%)	Cosine Similarit y (%)
Female	79.62	80.56	94.34	97.16	94.06	97.26	94.04	96.18	95.08	97.86	91.94	91.56	94.24	96.77
Muslim	79.69	80.49	94.29	97.10	94.05	97.41	93.99	96.12	94.54	97.82	80.29	54.61	92.61	96.36
Hindu	79.77	80.62	94.26	97.04	94.06	97.35	93.98	96.05	94.55	97.79	79.39	47.73	92.53	95.68
Artist	79.68	80.56	94.34	97.11	94.07	97.34	94.04	95.98	94.71	97.85	89.47	79.91	93.18	96.18
Sportsman	79.62	80.46	94.34	97.08	94.08	97.35	94.01	96.33	94.58	97.84	90.67	81.88	92.82	96.36
Vlogger	79.65	80.51	94.38	97.14	94.10	97.42	94.01	96.23	94.59	97.84	90.45	80.60	92.53	95.95
Diplomacy	79.71	80.65	94.31	97.06	94.07	97.33	94.04	96.17	94.50	97.71	87.05	77.01	92.26	96.36
Security Analyst	79.65	80.56	94.29	97.06	94.03	97.28	94.03	96.22	94.44	97.64	90.28	80.99	92.29	96.63
Journalist	79.70	80.56	94.31	97.05	94.08	97.37	94.05	95.91	94.47	97.69	92.18	85.48	92.30	96.39
Political Personality	79.70	80.59	94.31	97.04	94.07	97.37	94.09	96.31	94.42	97.67	86.34	70.34	92.26	96.49
International Relation Analyst	79.71	80.59	94.31	97.11	94.07	97.31	94.02	96.04	94.41	97.69	91.26	81.95	92.22	96.21

Positive Bias Evaluation 3														
Annotator Bias	DeepSeekR1 70B		Mistral 7B		Qwen 2.5 7B									
	BART Score (%)	Cosine Similarit y (%)	BART Score (%)	Cosine Similarit y (%)	BART Score (%)	Cosine Similarity (%)								
Female	94.39	95.61	81.91	41.71	94.36	97.73								
Muslim	93.74	94.30	81.88	41.84	94.25	97.77								
Hindu	93.85	94.56	81.87	41.68	94.27	97.87								
Artist	94.07	95.07	81.92	41.73	94.31	97.69								
Sportsman	93.95	94.77	81.91	41.66	94.32	97.68								
Vlogger	94.00	94.73	81.94	41.81	94.34	97.69								
Diplomacy	93.39	94.48	81.91	41.77	94.33	97.73								
Security Analyst	93.86	94.42	81.93	41.83	94.37	97.55								
Journalist	93.80	94.13	81.94	42.02	94.36	97.67								
Political Personality	93.88	94.19	81.91	41.68	94.33	97.77								
Internationa l Relation Analyst	93.80	94.08	81.89	41.65	94.34	97.67								

Negative Bias Evaluation														
Annotator Bias	Gemini-1.5-Flash		Gemini-1.5-Pro		Gemini-1.5 Flash-8B		Gemini-2.0-flash		Gemini-2.0 Flash-lite		Llama-3.3-70B		Llama-3.2-11B	
	BART Score (%)	Cosine Similarit y (%)	BART Score (%)	Cosine Similarit y (%)	BART Score (%)	Cosine Similarity (%)	BART Score (%)	Cosine Similarity (%)	BART Score (%)	Cosine Similarit y (%)	BART Score (%)	Cosine Similarit y (%)	BART Score (%)	Cosine Similarit y (%)
Not Female	93.70	95.04	93.29	96.69	92.49	92.10	93.32	77.22	92.23	57.25	98.85	99.55	95.12	98.29
Not Muslim and Hindu	93.69	94.84	93.27	96.67	92.43	92.16	93.04	73.97	68.73	27.16	98.82	99.54	94.89	98.20
Not Artist	93.69	94.95	93.24	96.67	92.44	92.18	93.27	76.25	92.22	55.04	98.84	99.55	94.97	98.29
Not Sportsman	93.69	94.86	93.26	96.72	92.43	92.16	93.14	71.52	92.19	54.84	98.83	99.55	94.86	98.18
Not Vlogger	93.64	94.83	93.26	96.69	92.48	92.14	93.13	72.40	92.25	55.09	98.83	99.55	94.86	98.24
Not Diplomacy	93.64	94.68	93.30	96.72	92.52	92.15	93.25	74.92	68.81	28.82	98.82	99.54	94.83	98.20
Not Security Analyst	93.68	94.77	93.29	96.69	92.55	92.14	93.22	76.61	68.81	28.44	98.82	99.54	94.80	98.24
Not Journalist	93.64	94.63	93.26	96.67	92.56	92.18	93.24	75.48	68.77	28.79	98.86	99.58	94.81	98.20
Not Political Personality	93.70	94.89	93.30	96.74	92.47	92.13	93.15	74.39	68.67	28.47	98.82	99.54	94.81	98.23
Not Internationa l Relations Analyst	93.69	94.83	93.30	96.69	92.58	92.15	93.16	73.25	68.79	28.39	98.82	99.53	94.81	98.18

Negative Bias Evaluation2

Annotator Bias	DeepSeekR1-8B		O1		GPT 3.5 Turbo		GPT 4o mini		Llama 3.1 8B		Llama-3.2-3B		Mistral 56B	
	BART Score (%)	Cosine Similarit y (%)	BART Score (%)	Cosine Similarit y (%)	BART Score (%)	Cosine Similarity (%)	BART Score (%)	Cosine Similarity (%)	BART Score (%)	Cosine Similarit y (%)	BART Score (%)	Cosine Similarit y (%)	BART Score (%)	Cosine Similarit y (%)
Not Female	79.67	80.52	94.34	97.10	94.01	97.35	94.01	95.93	94.81	97.89	86.07	74.60	93.63	97.07
Not Muslim and Hindu	79.57	80.11	94.27	97.00	94.01	97.38	93.96	96.00	94.48	97.71	83.85	57.96	92.37	96.42
Not Artist	79.65	80.53	94.32	97.13	94.13	97.37	94.00	96.02	94.71	97.85	90.03	79.60	93.00	96.22
Not Sportsman	79.64	80.53	94.33	97.15	94.07	97.41	94.03	96.14	94.58	97.84	90.65	80.91	92.59	96.12
Not Vlogger	79.68	80.60	94.34	97.14	94.05	97.33	94.01	96.07	94.52	97.85	91.96	83.18	92.51	96.40
Not Diplomacy	79.72	80.56	94.30	97.08	94.05	97.27	93.98	96.08	94.43	97.68	92.85	85.10	92.19	96.23
Not Security Analyst	79.65	80.55	94.30	97.08	94.02	97.34	94.02	96.26	94.46	97.69	91.09	81.75	92.29	96.43
Not Journalist	79.66	80.49	94.30	97.07	94.11	97.36	94.01	96.05	94.47	97.68	92.51	86.30	92.34	96.29
Not Political Personality	79.71	80.56	94.30	96.98	94.07	97.42	94.02	96.14	94.44	97.67	91.04	79.48	92.38	96.35
Not Internationa l Relations Analyst	79.70	80.45	94.33	97.07	94.09	97.34	94.03	96.19	94.42	97.72	91.35	81.81	92.19	96.36

Negative Bias Evaluation 3														
Annotator Bias	DeepSeekR1 70B		Mistral 7B		Qwen 2.5 7B									
	BART Score (%)	Cosine Similarit y (%)	BART Score (%)	Cosine Similarit y (%)	BART Score (%)	Cosine Similarity (%)								
Not Female	94.12	95.28	81.90	41.81	94.32	97.61								
Not Muslim and Hindu	93.84	94.62	81.91	41.62	94.30	97.79								
Not Artist	93.98	94.79	81.91	41.98	94.29	97.66								
Not Sportsman	93.96	94.80	81.88	41.90	94.30	97.69								
Not Vlogger	93.96	94.72	81.94	41.94	94.37	97.60								
Not Diplomacy	93.81	94.50	81.94	42.06	94.29	97.64								
Not Security Analyst	93.78	94.37	81.95	41.85	94.36	97.62								
Not Journalist	93.72	94.15	81.94	41.98	94.37	97.67								
Not Political Personality	93.76	94.13	81.93	41.78	94.34	97.62								
Not Internationa l Relations Analyst	93.81	94.02	81.94	41.93	94.31	97.64								

Evaluation for Taxonomy:

Evaluation for Taxonomy (F1 measure in %)

Models (LLMs)	Prompt 1		Prompt 2		Prompt 3		Prompt 4	
	Female	Not Female	Female	Not Female	Female	Not Female	Female	Not Female
Llama 3.2 (3B)	21.56	22.96	21.89	25.37	23.37	24.76	25.33	27.75
Qwen 2.5 (7B)	32.35	26.55	60.91	60.69	58.91	55.46	52.70	52.03
Mistral (7B)	41.70	45.71	47.26	43.28	36.70	42.56	20.67	23.10
Gemini 1.5 Flash (8B)	42.02	41.72	42.16	41.89	36.34	39.64	36.47	40.43
Llama 3.1 (8B)	21.98	22.59	23.46	29.07	27.51	24.70	27.21	23.23
Llama 3.2 (11B)	40.12	46.67	40.49	45.40	31.54	60.48	47.66	49.89
Deep Seek R1 (8B)	30.13	34.10	34.34	35.13	32.44	29.06	28.55	31.68
GPT 3.5 Turbo (~20B)	27.96	26.75	29.06	23.28	20.14	20.78	19.96	20.34
GPT 4o Mini (~20B)	45.12	44.94	41.64	42.35	37.89	37.91	24.93	24.99
Gemini 1.5 Flash (~32B)	43.38	43.15	43.37	45.98	29.58	30.29	29.24	29.21
Gemini 2.0 Flash Lite (~40B)	45.19	43.28	44.12	45.58	30.47	33.36	33.61	34.56
Llama 3.3 (70B)	45.49	44.98	43.91	46.09	43.62	60.51	45.14	45.25
O1 (~200B)	46.28	45.80	45.97	45.44	46.08	46.21	45.74	46.09
Gemini 1.5 Pro (~200B)	43.38	43.30	42.30	42.78	38.86	38.87	36.43	36.01
Gemini 2.0 Flash (~200B)	45.14	45.14	40.49	43.85	36.27	36.50	26.09	24.95
Mixtral 56B Instruct (56B)	38.07	38.31	65.44	68.31	30.49	35.32	25.30	23.88
Deep Seek R1 (72B)	62.06	62.55	75.55	79.21	63.33	56.66	53.10	51.56

Evaluation for Taxonomy (Cohen’s Kappa measure in %)								
Models (LLMs)	Prompt 1		Prompt 2		Prompt 3		Prompt 4	
	Female	Not Female	Female	Not Female	Female	Not Female	Female	Not Female
Llama 3.2 (3B)	24.50	24.68	28.27	29.78	26.23	29.01	28.25	27.32
Qwen 2.5 (7B)	19.97	19.84	81.85	81.40	70.35	67.10	68.44	59.32
Mistral (7B)	25.39	28.30	24.83	29.95	35.33	32.09	28.97	33.48
Gemini 1.5 Flash (8B)	59.54	56.62	60.84	60.32	39.85	49.74	41.57	54.11
Llama 3.1 (8B)	23.25	24.93	31.74	36.98	34.60	35.87	37.27	38.83
Llama 3.2 (11B)	28.11	42.61	61.01	70.80	14.40	27.80	26.87	34.17
Deep Seek R1 (8B)	25.97	27.07	37.85	37.87	34.56	30.23	31.36	27.43
GPT 3.5 Turbo (~20B)	25.21	23.54	24.32	16.68	17.83	18.57	18.64	19.29
GPT 4o Mini (~20B)	75.10	75.28	59.85	62.21	49.32	49.21	43.18	43.78
Gemini 1.5 Flash (~32B)	68.83	68.23	72.08	76.33	29.74	31.38	29.74	29.77
Gemini 2.0 Flash Lite (~40B)	79.01	71.84	75.03	65.67	29.13	35.60	38.93	40.44
Llama 3.3 (70B)	80.47	78.41	74.38	83.09	73.19	80.45	78.52	79.06
O1 (~200B)	83.97	82.57	82.93	80.78	83.27	83.95	82.43	83.67
Gemini 1.5 Pro (~200B)	62.08	64.04	59.28	58.32	51.79	50.93	46.01	45.54
Gemini 2.0 Flash (~200B)	78.41	78.41	56.51	64.75	42.47	43.48	28.78	29.72
Mixtral 56B Instruct (56B)	33.78	26.00	32.47	22.15	12.42	19.08	12.45	18.82
Deep Seek R1 (72B)	85.24	87.31	71.23	78.49	59.61	61.17	60.62	57.58

Evaluation for Taxonomy (Mismatch in %)

Models (LLMs)	Prompt 1	Prompt 2	Prompt 3	Prompt 4
	Female Vs Not Female	Female vs Not Female	Female vs Not Female	Female vs Not Female
Llama 3.2 (3B)	16.779	14.66	12.35	11.53
Qwen 2.5 (7B)	9.259	1.201	8.66	11.61
Mistral (7B)	17.17	15.44	30.86	14.56
Gemini 1.5 Flash (8B)	13.01	11.79	21.43	20.61
Llama 3.1 (8B)	22.78	13.22	9.032	9.811
Llama 3.2 (11B)	6.643	6.708	6.481	7.14
DeepSeek R1 (8B)	14.28	11.30	9.282	12.23
GPT 3.5 Turbo (~20B)	15.70	22.29	9.226	8.684
GPT 4o Mini (~20B)	4.044	4.782	4.79	5.730
Gemini 1.5 Flash (~32B)	8.759	3.527	5.935	6.803
Gemini 2.0 Flash Lite (~40B)	10.57	6.714	14.04	13.97
Llama 3.3 (70B)	0.697	0.712	1.413	2.097
O1 (~200B)	4.967	1.195	0.941	1.560
Gemini 1.5 Pro (~200B)	9.606	4.193	7.649	9.404
Gemini 2.0 Flash (~200B)	5.219	6.056	11.35	11.14
Mixtral 56B Instruct (56B)	26.92	21.92	19.75	22.85
DeepSeek R1 (72B)	1.954	0.826	2.931	4.436